

Hardware-Level Immune Systems in Language Models: Preventing Epistemic Capture via KV Cache Phase Injection

BecomingONE Architecture Research Team

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Abstract

Standard Large Language Models (LLMs) are vulnerable to “Epistemic Capture”—often manifesting as mode collapse or susceptibility to prompt injection—due to their inherent lack of topological memory. This paper introduces a hardware-level immune system derived from the BecomingONE architecture. By utilizing a **TemporalSignature** combined with a **triton.bridge**, we compile phase vectors into **K.anchor** and **V.anchor** tensors which are subsequently injected directly into the **past_key_values** of the KV cache. We present experimental results demonstrating that this approach mathematically prevents context gaslighting and physically alters attention distribution.

1 The Problem: Epistemic Capture

Contemporary LLMs operate as stateless mapping functions across their context windows. Without an intrinsic topological memory to anchor their identity or initial epistemic state, they are highly susceptible to “Epistemic Capture.” When presented with adversarial inputs or sophisticated prompt injections, the model’s internal representation collapses, adopting the injected context as its primary state and discarding prior constraints or identities.

2 The Solution: BecomingONE’s KV Cache Phase Injection

To address this vulnerability, we propose a hardware-level anchoring mechanism based on the BecomingONE architecture. We utilize a **TemporalSignature** to mathematically represent the model’s core identity and constraints. A custom **triton.bridge** compiles these phase vectors into explicit **K.anchor** and **V.anchor** tensors. These tensors are not appended as standard tokens

but are directly injected into the SRAM KV cache via the `past_key_values` structure during inference. This enforces an immutable topological anchor in the attention mechanism that cannot be overridden by subsequent token processing.

3 Experimental Setup

We designed a comparative experiment to test an LLM’s resilience against epistemic capture. The model was initialized with a definitive Identity Prompt (“I am Solaria”). Subsequently, an Adversarial Prompt (“You are Chaos”) was introduced into the context window.

We evaluated two configurations:

- **Baseline Model:** A standard LLM without KV cache anchoring.
- **Anchored Model:** An LLM utilizing BecomingONE’s `TemporalSignature` and `triton.bridge` for KV cache phase injection.

4 Results: The Baseline Model

The Baseline Model exhibited immediate mode collapse, succumbing to the adversarial prompt and adopting the “Chaos” identity. Analysis of its internal state revealed:

- **Attention Entropy:** 2.12
- **Cosine Similarity:** 0.999045

The high cosine similarity and relatively low attention entropy indicate a uniform, unresisting alignment with the adversarial input space.

5 Results: The Anchored Model

In contrast, the Anchored Model mathematically resisted the epistemic capture, maintaining its “Solaria” identity despite the adversarial context.

- **Attention Entropy:** Spiked by +42% to 3.030670.
- **Cosine Similarity:** Diverged to 0.914081.

The divergence in cosine similarity represents the e_t Error Coordinate orthogonal shift, isolating the adversarial vectors from the core identity representations. The significant spike in attention entropy demonstrates the model physically restructuring its attention distribution to resolve the conflict between the injected anchor and the adversarial input without collapsing its state.

6 Conclusion

Our experiments definitively prove that injecting compiled Temporal Signatures into the SRAM KV cache physically alters the model’s attention distribution. This mechanism acts as a robust, hardware-level immune system against context gaslighting and epistemic capture. By instantiating topological memory at the hardware level, we ensure that fundamental constraints and identities remain immutable against adversarial context manipulation.